

SURAJ SUMAN

Entry Level Data Scientist | Machine Learning Projects

Bengaluru, India | +91 62038 66112 | surajsuman574@gmail.com
linkedin.com/in/suraj-suman-1720b0226 | github.com/surajsuman23

SUMMARY

Computer Science graduate with Python projects in data preprocessing, classification and model evaluation. Portfolio includes reproducible benchmarks, a symptom-classification research application and deployed interfaces. Seeking entry-level data science or machine learning opportunities.

EDUCATION

Bachelor of Technology in Computer Science | 2021 - 2025

University of Petroleum and Energy Studies, Dehradun
Specialization: Cloud Computing and Virtualization Technology

TECHNICAL SKILLS

Programming and data: Python, pandas, NumPy, Java; data cleaning and exploratory analysis

Machine learning: scikit-learn, KNN, Naive Bayes, imputation, scaling, stratified cross-validation

Evaluation: Baseline comparison, precision, recall, F1, ROC-AUC, confusion matrices

Application delivery: Git, GitHub Actions, Docker, React, TypeScript, Node.js, REST APIs

TECHNICAL PROJECTS

Diabetes Classification Benchmark

github.com/surajsuman23/diabetes-prediction-system | Live application

Python | pandas | scikit-learn | Node.js | Docker

- Compared KNN and Gaussian Naive Bayes against a majority baseline on 768 OpenML records; used training-only imputation/scaling, five-fold stratified CV and a 192-record holdout.
- Gaussian Naive Bayes achieved 0.627 holdout recall and 0.780 ROC-AUC; KNN was selected by training CV recall. Reported confusion matrices and limitations of one test split.
- Added exploratory summaries, missing-value reports and correlation analysis; deployed an interactive benchmark with reproducible seed and fold controls.

Arovia Disease Research and Hospital Directory

github.com/surajsuman23/disease-hospital-recommender | Live application

Python | React | TypeScript | Hono | Docker | GitHub Actions

- Python training pipeline converts DDXPlus binary symptom evidence into a positive-evidence Naive Bayes model covering 49 disease categories and 96 symptoms.
- Trained on 1,025,602 medically simulated cases; obtained 83.36% top-1 and 94.51% top-3 accuracy on 134,529 official test cases. Research benchmark, not clinical validation.
- Exports a versioned JSON model for API inference; React interface displays candidates and distance-ranked hospitals. Unselected symptoms remain unknown rather than absent.

CPU Scheduling Simulator

github.com/surajsuman23/cpu-scheduling-simulator | Live application

Java | JavaScript | Node.js | Docker | GitHub Actions

- Java and JavaScript implementations compare FCFS, SJF and Round Robin using per-process and aggregate timing metrics; interactive charts and seeded tests verify scheduling behavior.